B

I Dewa Nyoman Bayu Satria Wibawa

`Machine Learning & Software Engineer`

idewanyomanbayusw.vercel.app | idewanyomanbayusw@gmail.com | github.com/idewbayu14



AVAILABLE FOR ML & SOFTWARE ENGINEERING ROLES

Engineering Portfolio & Applied Research

Machine Learning Engineer specializing in Natural Language Processing, text classification pipelines, and production inference systems. Experienced in end-to-end model development at the National Research and Innovation Agency (BRIN), algorithmic resampling research, and real-time telemetry dashboards.

1. NLP & TEXT CLASSIFICATION

Sentiment analysis, deep learning text classification, domain tokenization, lemmatization, and social opinion monitoring.

2. INTELLIGENT RECOMMENDERS

Multi-criteria constraint mapping, hybrid feature matching, and user state evaluation across curated data profiles.

3. PRODUCTION INFERENCE

PyTorch model serving, low-latency REST APIs, full-stack React dashboards, and high-throughput IoT telemetry.

Technical Competencies & Systems Ecosystem

Core Methodologies

Machine Learning & Deep Learning

Python **PyTorch** **NLP**

Scikit-Learn | Transformers | Sentiment Analysis | Recommender Systems | Deep Learning | Data Mining

Data Pipelines & Feature Engineering

Pandas **NumPy**

TF-IDF Vectorization | Tokenization | Lemmatization | Web Scraping | Data Preprocessing | ROC-AUC Evaluation

Production, Systems & Cloud

FastAPI **React** **TypeScript**

Next.js | RESTful APIs | Tailwind CSS | Docker | Git & GitHub | Vercel | Linux Shell

FEATURED ENGINEERING DELIVERABLE • 01

BRIN Research 2026 `brin-project-kp.vercel.app`

Autonomous Vehicle Sentiment Analysis Platform

Key Research Highlight: End-to-end deep learning NLP platform assessing consumer acceptance and readiness for Level 2 autonomous vehicles. Engineered with an independently collected research dataset and a self-trained multi-class neural classification model achieving **94.2% accuracy**, deployed as a self-hosted live inference service without external third-party black-box models. *(Detailed breakdown on Page 2)*

PROJECT 01 • NATIONAL RESEARCH AND INNOVATION AGENCY (BRIN) • 2026

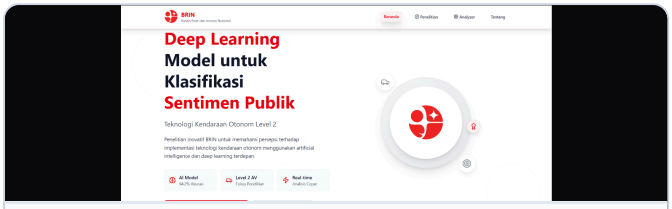
Machine Learning & NLP `brin-project-kp.vercel.app`

Autonomous Vehicle Sentiment Analysis Platform

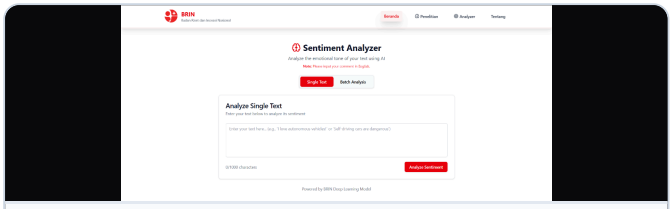
End-to-end deep learning natural language processing platform analyzing unstructured public discussions across YouTube and Reddit to assess consumer acceptance and readiness for Level 2 autonomous vehicles. Built from raw data ingestion to live cloud production serving.

- **Independent Research Dataset:** Scraped, curated, labeled, and preprocessed raw public discussion datasets with custom Indonesian and domain-specific text normalization.
- **Self-Trained Deep Learning Model:** Architected, trained, and evaluated a multi-class deep learning text classification model reaching **94.2% accuracy** across positive, neutral, and negative sentiment classes without reliance on third-party black-box APIs.
- **End-to-End Production Deployment:** Engineered a low-latency REST inference pipeline integrated directly into a responsive, live research dashboard hosted on Vercel.

Architecture: Raw Public Discussions → Automated Ingestion → Custom Tokenization & Lemmatization → Self-Trained Deep Learning Classifier → FastAPI / REST Inference API → Live React Research Platform.



Research Portal & In-House Deep Learning Classification Architecture



Live Interactive Real-Time Sentiment Analyzer (Single & Batch Analysis)

Python | Deep Learning | NLP | Custom Dataset | PyTorch | REST API | React | Web Scraping

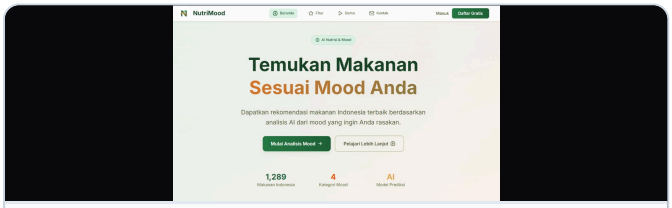
PROJECT 02 • DICODING & DBS FOUNDATION CAPSTONE • 2025

Machine Learning & AI `nutrimoodv.vercel.app`

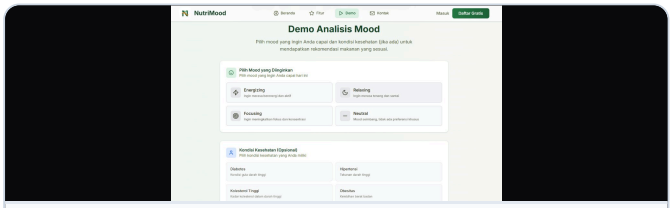
NutriMood: Personalized Dietary Recommender

Intelligent dietary recommendation web application translating user emotional states and personal medical restrictions into tailored daily meal recommendations.

- **Multi-Criteria Algorithm:** Engineered personalized recommendation algorithms mapping psychological mood states to nutritional requirements and caloric constraints.
- **Curated Culinary Knowledge Base:** Constructed end-to-end data preprocessing and nutritional feature-matching pipelines across 1,200+ culinary items.
- **Production Serving:** Packaged and deployed the trained recommender model as a cloud REST API integrated into a responsive web application.



NutriMood Application Homepage with AI-Driven Meal Recommendation Engine



Interactive Mood & Medical Health Constraint Evaluation Interface

Machine Learning | Recommender System | Python | Feature Matching | Data Preprocessing | REST API | React Web App

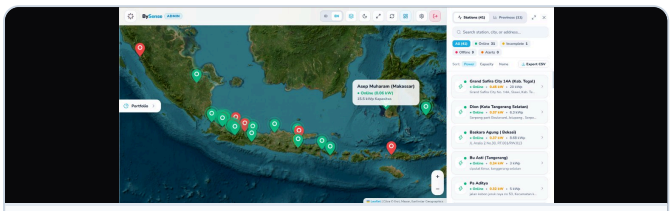
PROJECT 03 • BATARI ENERGY R&D ENGINEERING • 2026

BySense: Photovoltaic & Solar IoT Telemetry

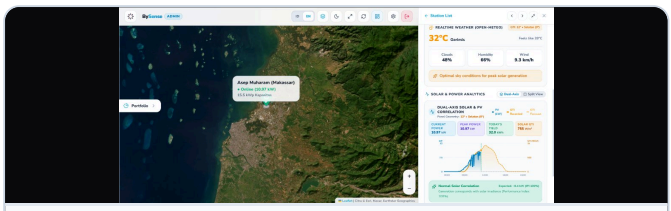
IoT & Full-Stack System `bysense.batarienergy.com`

Cross-platform IoT telemetry platform monitoring real-time photovoltaic power generation, solar radiation forecasts, and energy flow across nationwide commercial installations.

- **Full-Stack Architecture:** Developed responsive web telemetry dashboard and React Native mobile app from scratch with live generation yield curves.
- **Environmental Streams & Analytics:** Integrated solar irradiance feeds (Open-Meteo) and inverter APIs for anomaly detection, energy forecasting, and dual-axis correlation.
- **Mobile Store Release:** Completed performance optimization, edge debugging, and production release to the Google Play Store.



Nationwide Telemetry Map of Operational Solar Power Plants



Dual-Axis Solar & PV Generation Correlation Analytics

React | Next.js | React Native | REST API | IoT Telemetry | Data Visualization | Mobile Optimization

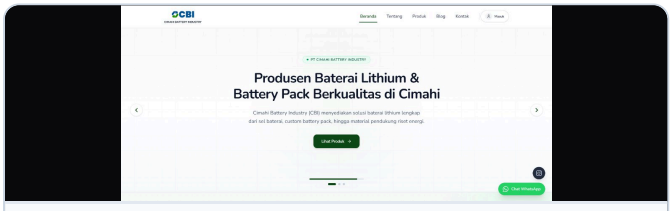
PROJECT 04 • PRODUCTION INDUSTRIAL WEB DEPLOYMENT • 2026

Cimahi Battery Commercial Web Platform

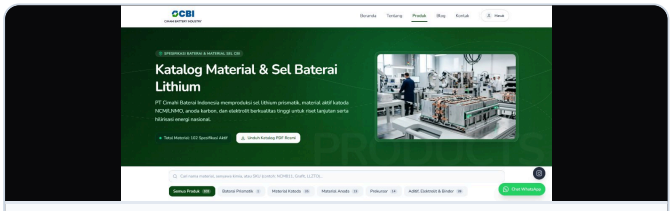
Industrial Commercial Web `cimahibattery.com`

High-performance commercial web platform built for industrial lithium battery manufacturers, featuring structured prismatic cell catalogs and active energy material indexes.

- **Structured Catalog Architecture:** Designed and deployed a responsive commercial platform featuring structured battery pack categorization and instant search filtering.
- **Performance & SEO Optimization:** Optimized Core Web Vitals, accessibility compliance, and search visibility for national industrial energy procurement.



Official Commercial Homepage of PT Cimahi Battery Industry



Structured Prismatic Battery & Active Energy Material Index

Web Development | Responsive UI | Frontend Architecture | SEO Optimization | Production Hosting

Digital Portfolio & Contact Verification

All projects, live deployments, and source repositories are verified and available online.

Website: `https://idewanyomanbayusw.vercel.app/`
Email: idewanyomanbayusw@gmail.com • LinkedIn: in/idewanyomanbayu